

Practical no. 26

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include <stdio.h>
#include <stdlib.h>
#include <conio.h>
struct Node {
    int data;
    struct Node* left;
    struct Node* right;
};
struct Node* createNode(int value) {
    struct Node* newNode = (struct Node*)malloc(sizeof(struct Node));
    newNode->data = value;
    newNode->left = newNode->right = NULL;
    return newNode;
}
struct Node* insert(struct Node* root, int value) {
    if (root == NULL) {
        return createNode(value);
    }
    if (value < root->data) {
        root->left = insert(root->left, value); // insert in left subtree
    }
    else if (value > root->data) {
        root->right = insert(root->right, value); // insert in right subtree
    }
    return root;
}
void inorder(struct Node* root) {
    if (root == NULL) return;
    inorder(root->left);
    printf("%d ", root->data);
    inorder(root->right);
}
void main() {
    struct Node* root = NULL;
    int n, value, i;
```

```
printf("\nEnter number of nodes: ");
scanf("%d", &n);
for (i = 0; i < n; i++) {
    printf("\nEnter value for node %d: ", i + 1);
    scanf("%d", &value);
    root = insert(root, value);
}
printf("\nInorder Traversal of BST: ");
inorder(root);
printf("\n");
getch();
}
```

OUTPUT :-

```
Enter number of nodes: 4
Enter value for node 1: 10
Enter value for node 2: 20
Enter value for node 3: 30
Enter value for node 4: 40
Inorder Traversal of BST: 10 20 30 40
```